



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Mid Term-I (Session: 2025-26)

Roll No.:

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Course Name: B.tech
Subject Name: DS
Subject Code: 25CS101

Max. Marks: 15
Time: 75 Minutes
Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions		(1 x 5)		
(a)	Draw Memory hierarchy	CO1	K2	2.1.1
(b)	Example of syntax and semantic error	CO1	K2	2.1.1
(c)	Define the characteristics of an Algorithm	CO1	K2	2.1.2
(d)	Find the value of $z = !(10 * 0)$ in the expression	CO2	K2	2.1.1
(e)	Find the value of $z = ++x + ++x$, where $x = 10$	CO2	K2	2.1.1
SECTION-B (Minimum Knowledge Level: K2)				
2. Attempt Any two Question		(Same K Levels) (2 x 2)		
(a)	Difference between compiler and interpreter	CO1	K1	2.1.2
(b)	Draw flow chart for temperature conversion from Fahrenheit to Celsius(hint: $-f - 32 = 9c / 5$)	CO1	K1	2.1.1
(c)	Difference between Multiprogramming and Multitasking	CO1	K1	1.4.1

Solutions for Mid Term-I (Session: 2025-26)

Subject: Data Structures (25CS101)

Semester: I

Max Marks: 15

Time: 75 Minutes

SECTION-A (Maximum Knowledge Level: K2)

1. 1. Attempt All Questions (1 x 5)

(a) Memory hierarchy:

Registers → Cache → Main Memory (RAM) → Secondary Memory (Hard Disk) → Tertiary Storage (Optical Disk/Tape).

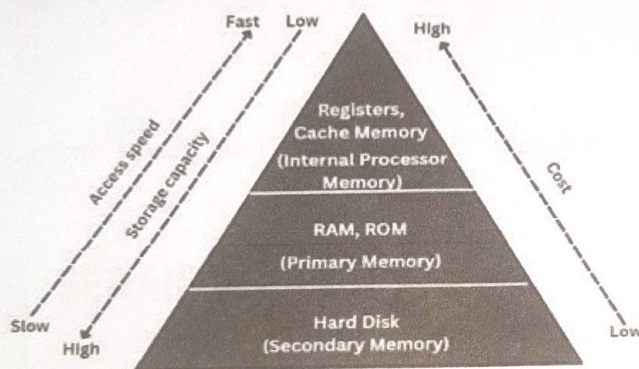


Fig: Memory Hierarchy

(b) Example of syntax and semantic error:

Example of Syntax Error:

```
#inclu  
de  
<iostr  
eam>  
using  
namesp
```

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```
ace  
std;
```

```
int main() {  
    int num  
    cout << "Enter a number: "; // Missing  
    semicolon above  
    cin >> num;  
    cout << "You entered: "  
    << num; return 0;  
}
```

Error: Missing semicolon after 'int num'. The compiler will give an error message like: error: expected ';' before 'cout'

Corrected Code:

```
int num;
```

Syntax Error: Missing semicolon after a statement (e.g., int x = 10).

SEMANTIC ERROR

- It occurs when the statements written in the program are not meaningful to the compiler.
- Difficult to detect because their syntax is correct but their meaning is wrong.

Use of a non-initialized variable

Wrong statement

Type incompatibility

```
//C++ program to illustrate semantic error  
#include<iostream.h>  
int main ()  
{  
    int i;  
    i++; //Error: The variable 'i' is not  
    initialized  
}
```

Semantic Error: Using an uninitialized variable (e.g., `int x; y = x + 5;`).

(c) Characteristics of an Algorithm:

- a. **Finiteness** – must end after finite steps.
- b. **Definiteness** – each step must be clear.
- c. **Input** – takes zero or more inputs.
- d. **Output** – produces at least one output.
- e. **Effectiveness** – steps must be basic and doable

(d) Find the value of $z = !(10 * 0)$:

$10 * 0 = 0 \rightarrow !(0) = 1 \rightarrow$ Hence, $z = 1$.

(e) Find the value of $z = ++x + ++x$, where $x = 10$:

After evaluation: $x = 10 + 12$ implies $x = 23$

OR

After first increment: $x = 11 \rightarrow$ second increment: $x = 12 \rightarrow z = 12 + 12$ implies $x = 24$.

OR

Compiler Dependent

SECTION-B (Minimum Knowledge Level: K2)

2. Attempt Any Two Questions (2 x 2)

(a) Difference between Compiler and Interpreter:

Compiler	Interpreter
Translates the entire program before execution.	Translates and executes line by line.
Execution is faster.	Execution is slower.

(b) Flowchart for temperature conversion from Fahrenheit to Celsius:

Start $\rightarrow$ Input Fahrenheit (F) $\rightarrow$ Compute $C = (F - 32) * 5/9 \rightarrow$ Display C $\rightarrow$ Stop.

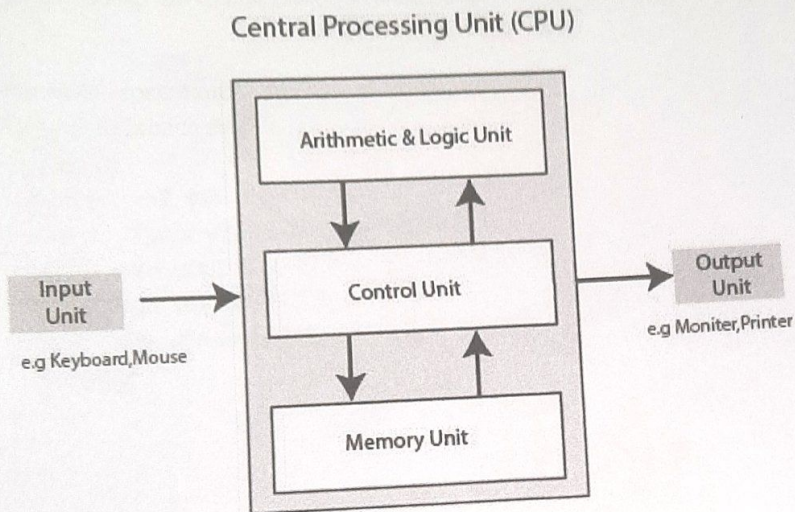
(c) Difference between Multiprogramming and Multitasking:

Multiprogramming	Multitasking
Multiple programs loaded in memory at the same time.	Multiple tasks executed by CPU seemingly simultaneously.
Increases CPU utilization.	Improves user interactivity.

SECTION-C (Minimum Knowledge Level: K3)

3. Attempt Any One Question (3 x 1)

(a) Block diagram of computer and explanation:



Components:

1. Input Unit
2. Output Unit
3. Memory Unit
4. Control Unit
5. Arithmetic and Logic Unit (ALU)

Explanation: Input unit takes data; control unit manages operations; ALU performs calculations; memory stores data and results; output unit displays results.

(b) Why Operating System is known as Resource Manager:

An Operating System manages computer resources like CPU, memory, I/O devices, and files. It allocates and deallocates resources efficiently to ensure smooth execution.

4. Attempt Any One Question (3 x 1)

(a) C++ Program to check even or odd using ternary operator:

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;
```

```
(num % 2 == 0) ? cout << "Even" : cout << "Odd";  
return 0;  
}
```

(b) C++ Program to calculate average marks of three subjects:

```
#include <iostream>  
using namespace std;  
int main() {  
    float m1, m2, m3, avg;  
    cout << "Enter marks of three subjects: ";  
    cin >> m1 >> m2 >> m3;  
    avg = (m1 + m2 + m3) / 3;  
    cout << "Average Marks = " << avg;  
    return 0;  
}
```